```
import pandas as pd
```

```
df = pd.read_csv('/content/Heart_Disease_Prediction.csv')
```

```
df.head()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
0	70	1	4	130	322	0	2	109	0	2.4	2	3	3	Presence
1	67	0	3	115	564	0	2	160	0	1.6	2	0	7	Absence
2	57	1	2	124	261	0	0	141	0	0.3	1	0	7	Presence
3	64	1	4	128	263	0	0	105	1	0.2	2	1	7	Absence

```
df.tail()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Heart Disease
265	52	1	3	172	199	1	0	162	0	0.5	1	0	7	Absence
266	44	1	2	120	263	0	0	173	0	0.0	1	0	7	Absence
267	56	0	2	140	294	0	2	153	0	1.3	2	0	3	Absence

```
df.shape
```

```
(270, 14)
```

```
df.columns
```

```
Index(['Age', 'Sex', 'Chest pain type', 'BP', 'Cholesterol', 'FBS over 120',
       'EKG results', 'Max HR', 'Exercise angina', 'ST depression',
       'Slope of ST', 'Number of vessels fluro', 'Thallium', 'Heart Disease'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 270 entries, 0 to 269
Data columns (total 14 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       270 non-null    int64
 1   Sex                       270 non-null    int64
 2   Chest pain type           270 non-null    int64
 3   BP                        270 non-null    int64
 4   Cholesterol               270 non-null    int64
 5   FBS over 120              270 non-null    int64
 6   EKG results               270 non-null    int64
 7   Max HR                    270 non-null    int64
 8   Exercise angina           270 non-null    int64
 9   ST depression             270 non-null    float64
 10  Slope of ST               270 non-null    int64
 11  Number of vessels fluro   270 non-null    int64
 12  Thallium                  270 non-null    int64
 13  Heart Disease             270 non-null    object
dtypes: float64(1), int64(12), object(1)
memory usage: 29.7+ KB
```

```
df.dtypes
```

	0
Age	int64
Sex	int64
Chest pain type	int64
BP	int64
Cholesterol	int64
FBS over 120	int64
EKG results	int64
Max HR	int64
Exercise angina	int64
ST depression	float64
Slope of ST	int64
Number of vessels fluro	int64
Thallium	int64
Heart Disease	object

dtype: object

```
numerical_features = df.select_dtypes(include=['int64','float64'])
numerical_features.columns
```

```
Index(['Age', 'Sex', 'Chest pain type', 'BP', 'Cholesterol', 'FBS over 120',
       'EKG results', 'Max HR', 'Exercise angina', 'ST depression',
       'Slope of ST', 'Number of vessels fluro', 'Thallium'],
      dtype='object')
```

```
categorical_features = df.select_dtypes(include=['object'])
categorical_features.columns
```

```
Index(['Heart Disease'], dtype='object')
```

```
df.isnull().sum()
```

	0
Age	0
Sex	0
Chest pain type	0
BP	0
Cholesterol	0
FBS over 120	0
EKG results	0
Max HR	0
Exercise angina	0
ST depression	0
Slope of ST	0
Number of vessels fluro	0
Thallium	0
Heart Disease	0

dtype: int64

```
df.isnull().values.any()
```

```
np.False_
```

```
df[df.isnull().any(axis=1)]
```

Age	Sex	Chest pain	BP	Cholesterol	FBS over	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels	Thallium	Heart Disease

```
df.describe()
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression
count	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.000000	270.00000
mean	54.433333	0.677778	3.174074	131.344444	249.659259	0.148148	1.022222	149.677778	0.329630	1.05000
std	9.109067	0.468195	0.950090	17.861608	51.686237	0.355906	0.997891	23.165717	0.470952	1.14521
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000	0.000000	71.000000	0.000000	0.00000
25%	48.000000	0.000000	3.000000	120.000000	213.000000	0.000000	0.000000	133.000000	0.000000	0.00000
50%	55.000000	1.000000	3.000000	130.000000	245.000000	0.000000	2.000000	153.500000	0.000000	0.80000
75%	61.000000	1.000000	4.000000	140.000000	280.000000	0.000000	2.000000	166.000000	1.000000	1.60000

```
import numpy as np
```

```
age_array=df['Age'].to_numpy()
age_array
```

```
array([70, 67, 57, 64, 74, 65, 56, 59, 60, 63, 59, 53, 44, 61, 57, 71, 46,
       53, 64, 40, 67, 48, 43, 47, 54, 48, 46, 51, 58, 71, 57, 66, 37, 59,
       50, 48, 61, 59, 42, 48, 40, 62, 44, 46, 59, 58, 49, 44, 66, 65, 42,
       52, 65, 63, 45, 41, 61, 60, 59, 62, 57, 51, 44, 60, 63, 57, 51, 58,
       44, 47, 61, 57, 70, 76, 67, 45, 45, 39, 42, 56, 58, 35, 58, 41, 57,
       42, 62, 59, 41, 50, 59, 61, 54, 54, 52, 47, 66, 58, 64, 50, 44, 67,
       49, 57, 63, 48, 51, 60, 59, 45, 55, 41, 60, 54, 42, 49, 46, 56, 66,
       56, 49, 54, 57, 65, 54, 54, 62, 52, 52, 60, 63, 66, 42, 64, 54, 46,
       67, 56, 34, 57, 64, 59, 50, 51, 54, 53, 52, 40, 58, 41, 41, 50, 54,
       64, 51, 46, 55, 45, 56, 66, 38, 62, 55, 58, 43, 64, 50, 53, 45, 65,
       69, 69, 67, 68, 34, 62, 51, 46, 67, 50, 42, 56, 41, 42, 53, 43, 56,
       52, 62, 70, 54, 70, 54, 35, 48, 55, 58, 54, 69, 77, 68, 58, 60, 51,
       55, 52, 60, 58, 64, 37, 59, 51, 43, 58, 29, 41, 63, 51, 54, 44, 54,
       65, 57, 63, 35, 41, 62, 43, 58, 52, 61, 39, 45, 52, 62, 62, 53, 43,
       47, 52, 68, 39, 53, 62, 51, 60, 65, 65, 60, 60, 54, 44, 44, 51, 59,
       71, 61, 55, 64, 43, 58, 60, 58, 49, 48, 52, 44, 56, 57, 67])
```

```
mean_age=np.mean(age_array)
median_age=np.median(age_array)
std_age=np.std(age_array)
print(mean_age, ',',median_age, ',',std_age)
```

```
54.43333333333333 , 55.0 , 9.092182234083177
```

```
Cholesterol_array = df['Cholesterol'].to_numpy()
Cholesterol_array
```

```
array([322, 564, 261, 263, 269, 177, 256, 239, 293, 407, 234, 226, 235,
       234, 303, 149, 311, 203, 211, 199, 229, 245, 303, 204, 288, 275,
       243, 295, 230, 265, 229, 228, 215, 326, 200, 256, 207, 273, 180,
       222, 223, 209, 233, 197, 218, 211, 149, 197, 246, 225, 315, 205,
       417, 195, 234, 198, 166, 178, 249, 281, 126, 305, 226, 240, 233,
       276, 261, 319, 242, 243, 260, 354, 245, 197, 223, 309, 208, 199,
       209, 236, 218, 198, 270, 214, 201, 244, 208, 270, 306, 243, 221,
       330, 266, 206, 212, 275, 302, 234, 313, 244, 141, 237, 269, 289,
       254, 274, 222, 258, 177, 160, 327, 235, 305, 304, 295, 271, 249,
       288, 226, 283, 188, 286, 274, 360, 273, 201, 267, 196, 201, 230,
       269, 212, 226, 246, 232, 177, 277, 249, 210, 207, 212, 271, 233,
       213, 283, 282, 230, 167, 224, 268, 250, 219, 267, 303, 256, 204,
       217, 308, 193, 228, 231, 244, 262, 259, 211, 325, 254, 197, 236,
       282, 234, 254, 299, 211, 182, 294, 298, 231, 254, 196, 240, 409,
       172, 265, 246, 315, 184, 233, 394, 269, 239, 174, 309, 282, 255,
       250, 248, 214, 239, 304, 277, 300, 258, 299, 289, 298, 318, 240,
       309, 250, 288, 245, 213, 216, 204, 204, 252, 227, 258, 220, 239,
       254, 168, 330, 183, 203, 263, 341, 283, 186, 307, 219, 260, 255,
       231, 164, 234, 177, 257, 325, 274, 321, 264, 268, 308, 253, 248,
       269, 185, 282, 188, 219, 290, 175, 212, 302, 243, 353, 335, 247,
       340, 206, 284, 266, 229, 199, 263, 294, 192, 286])
```

```
np.max(Cholesterol_array)
```

```
np.int64(564)
```

```
np.min(Cholesterol_array)
```

```
np.int64(126)
```

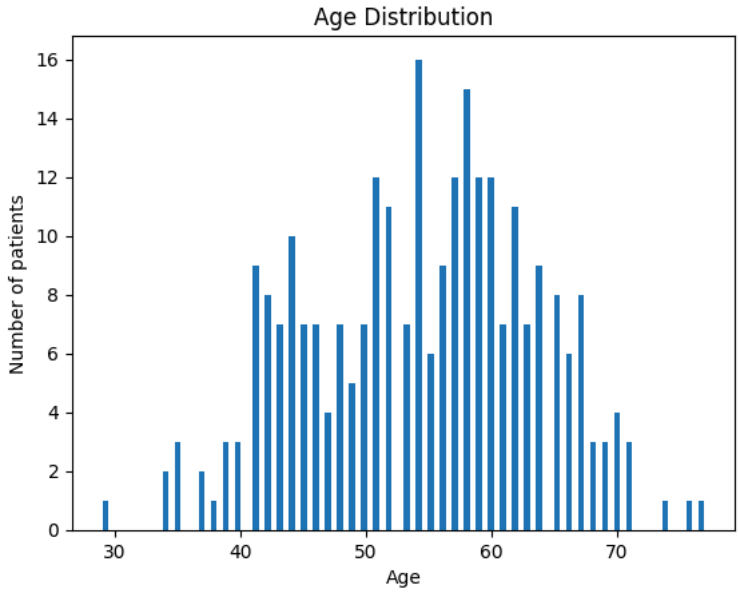
```
count_hd = df['Heart Disease'].value_counts()
count_hd
```

Heart Disease	count
Absence	150
Presence	120

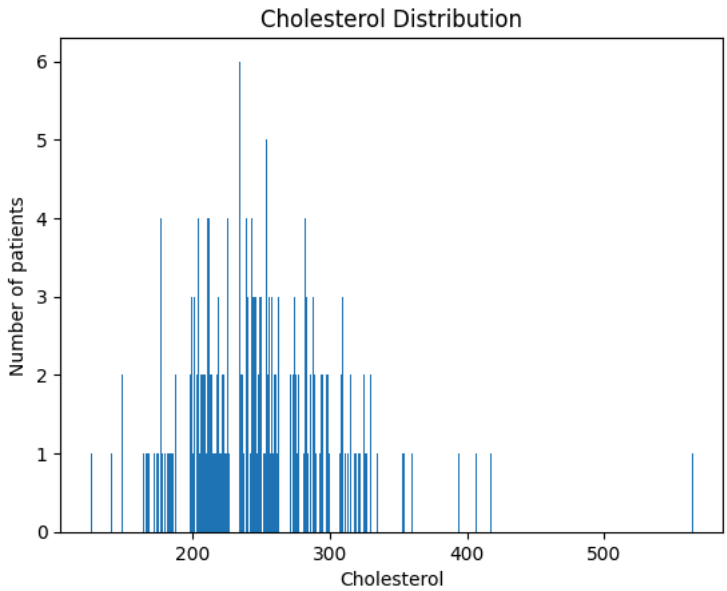
dtype: int64

```
import matplotlib.pyplot as plt
```

```
plt.hist(age_array, bins=100)
plt.xlabel('Age')
plt.ylabel('Number of patients')
plt.title('Age Distribution')
plt.show()
```

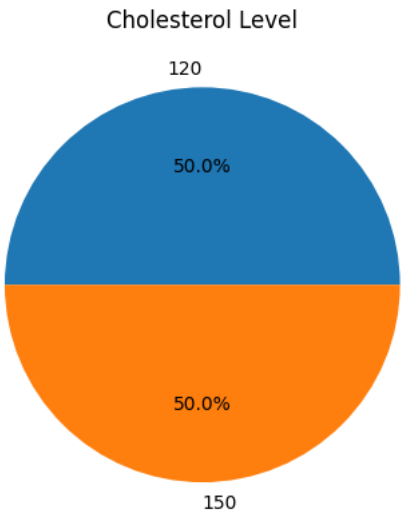



```
values, counts = np.unique(Cholesterol_array, return_counts=True)
plt.bar(values, counts)
plt.xlabel('Cholesterol')
plt.ylabel('Number of patients')
plt.title('Cholesterol Distribution')
plt.show()
```

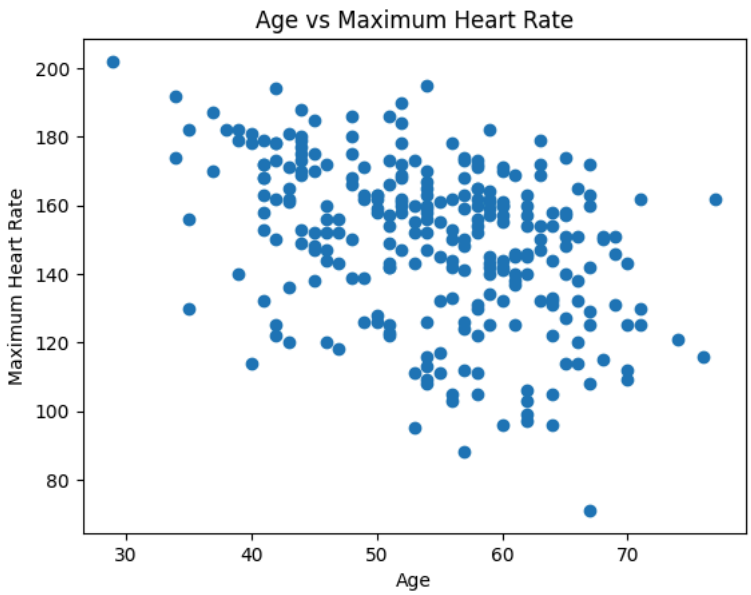



```
values, counts = np.unique(count_hd, return_counts=True)
plt.pie(counts, labels=values, autopct='%1.1f%%')
```

```
plt.title('Cholesterol Level')
plt.show()
```




```
plt.scatter(df['Age'], df['Max HR'])
plt.xlabel('Age')
plt.ylabel('Maximum Heart Rate')
plt.title('Age vs Maximum Heart Rate')
plt.show()
```




```
x=df.drop(columns='Heart Disease',axis=1)
y=df['Heart Disease']
correlation_matrix = df.corr()
correlation_matrix
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope
Age	1.000000	-0.094401	0.096920	0.273053	0.220056	0.123458	0.128171	-0.402215	0.098297	0.194234	0.1597
Sex	-0.094401	1.000000	0.034636	-0.062693	-0.201647	0.042140	0.039253	-0.076101	0.180022	0.097412	0.0505
Chest pain type	0.096920	0.034636	1.000000	-0.043196	0.090465	-0.098537	0.074325	-0.317682	0.353160	0.167244	0.1369
BP	0.273053	-0.062693	-0.043196	1.000000	0.173019	0.155681	0.116157	-0.039136	0.082793	0.222800	0.1424
Cholesterol	0.220056	-0.201647	0.090465	0.173019	1.000000	0.025186	0.167652	-0.018739	0.078243	0.027709	-0.0057
FBS over 120	0.123458	0.042140	-0.098537	0.155681	0.025186	1.000000	0.053499	0.022494	-0.004107	-0.025538	0.0440
EKG results	0.128171	0.039253	0.074325	0.116157	0.167652	0.053499	1.000000	-0.074628	0.095098	0.120034	0.1606
Max HR	-0.402215	-0.076101	-0.317682	-0.039136	-0.018739	0.022494	-0.074628	1.000000	-0.380719	-0.349045	-0.3868
Exercise angina	0.098297	0.180022	0.353160	0.082793	0.078243	-0.004107	0.095098	-0.380719	1.000000	0.274672	0.2559
ST depression	0.194234	0.097412	0.167244	0.222800	0.027709	-0.025538	0.120034	-0.349045	0.274672	1.000000	0.6097

```
correlation_age_max_hr = correlation_matrix.loc['Age', 'Max HR']
print(f"Correlation between Age and Max HR: {correlation_age_max_hr}")
```

```
Correlation between Age and Max HR: -0.40221544605675813
```

```
correlation_matrix.round(2)
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	Hea Diseas
Age	1.00	-0.09	0.10	0.27	0.22	0.12	0.13	-0.40	0.10	0.19	0.16	0.36	0.11	0.2
Sex	-0.09	1.00	0.03	-0.06	-0.20	0.04	0.04	-0.08	0.18	0.10	0.05	0.09	0.39	0.3
Chest pain type	0.10	0.03	1.00	-0.04	0.09	-0.10	0.07	-0.32	0.35	0.17	0.14	0.23	0.26	0.4
BP	0.27	-0.06	-0.04	1.00	0.17	0.16	0.12	-0.04	0.08	0.22	0.14	0.09	0.13	0.1
Cholesterol	0.22	-0.20	0.09	0.17	1.00	0.03	0.17	-0.02	0.08	0.03	-0.01	0.13	0.03	0.1
FBS over 120	0.12	0.04	-0.10	0.16	0.03	1.00	0.05	0.02	-0.00	-0.03	0.04	0.12	0.05	-0.0
EKG results	0.13	0.04	0.07	0.12	0.17	0.05	1.00	-0.07	0.10	0.12	0.16	0.11	0.01	0.1
Max HR	-0.40	-0.08	-0.32	-0.04	-0.02	0.02	-0.07	1.00	-0.38	-0.35	-0.39	-0.27	-0.25	-0.4
Exercise angina	0.10	0.18	0.35	0.08	0.08	-0.00	0.10	-0.38	1.00	0.27	0.26	0.15	0.32	0.4
ST depression	0.19	0.10	0.17	0.22	0.03	-0.03	0.12	-0.35	0.27	1.00	0.61	0.26	0.32	0.4

```
high_corr = correlation_matrix.abs() > 0.8
high_corr
```

	Age	Sex	Chest pain type	BP	Cholesterol	FBS over 120	EKG results	Max HR	Exercise angina	ST depression	Slope of ST	Number of vessels fluro	Thallium	He Dise
Age	True	False	False	False	False	False	False	False	False	False	False	False	False	F
Sex	False	True	False	False	False	False	False	False	False	False	False	False	False	F
Chest pain type	False	False	True	False	False	False	False	False	False	False	False	False	False	F
BP	False	False	False	True	False	False	False	False	False	False	False	False	False	F
Cholesterol	False	False	False	False	True	False	False	False	False	False	False	False	False	F
FBS over 120	False	False	False	False	False	True	False	False	False	False	False	False	False	F
EKG results	False	False	False	False	False	False	True	False	False	False	False	False	False	F
Max HR	False	False	False	False	False	False	False	True	False	False	False	False	False	F
Exercise angina	False	False	False	False	False	False	False	False	True	False	False	False	False	F
ST	False	False	False	False	False	False	False	False	False	True	False	False	False	F